

ANEMIA WARD

ANEMIA OF CKD

UNPLUGGED

- A short-life RBC
- DOWN RBC SURVIVAL
- HIGH HEPICIDIN
- DOWN Fe
- FUNCTIONAL Fe DEFICIENCY
- HYPERPTH
- MARROW FIBROSIS
- HEMOLYSIS
- B12 + FOLATE
- RBC
- NORMOCYTIC NORMOCHROMIC
- ANEMIA-MECHANISM ICONS
- DOWN EPO
- MAIN CAUSE + STARTS STAGE 3 + UNIVERSAL STAGE 4

HB TARGET 10 TO 11.5

DONT NORMALIZE = STROKE + THROMBOSIS

ANEMIA Rx

- ESA (EPOETIN ALFA)
- IRON IV
- TSAT
- FERRITIN
- ROXA + VADA + DAPRO + MOLIBDUSTAT
- TSAT + FERRITIN MUST BE OK FIRST
- ESA RESISTANCE
- INT. INFLAMMATION
- INADEQUATE DOSE
- HYPERPTH
- HEMOLYSIS
- INFECTION
- MALIGNANCY
- BLACKS PREFER HYDROXYUREA
- LIP Hb → LIP EPO → #

UREMIC BLEEDING

DDAVP

FIRST-LINE ACUTE vWF RELEASE + TACHYPHYLAXIS

CRYOPRECIPITATE vWF + FACTOR VIII

IV ESTROGENS + SUSTAINED SEVERE BLEEDING

DEFINITE ALSO HELPS PUSH PLATELETS TO WALL

TXA/EACA

DOWN AGGREGATION + DOWN PF3 + DOWN THROMBIN + LONG BLEEDING TIME

HYPERCOAGULABLE

ANTICOAGULATION

- Warfarin
- RIP
- LWH
- APIXABAN
- UFH
- heparin
- AVOID + CALCIPHYLAXIS RISK
- RENAL CLEARANCE AVOID
- PREFERRED DOAC
- SAFER MONITOR aPTT

NEPHROTIC SYNDROME = LOST ANTICOAG PROTEINS

ANEMIA WARD

PERIPHERAL NEUROPATHY

SENSORY > MOTOR

DISTAL > PROXIMAL

LEGS > ARMS

RESTLESS LEGS RELIEVED BY MOVEMENT

DIALYSIS INDICATION

ENDOCRINE APOTHECARY

IMPROVES WITH DIALYSIS/TX

WOMEN:

- DOWN ESTROGEN
- INFERTILITY
- GFR < 40
- ONLY 20% LIVE BIRTH

MEN:

- DOWN T
- SEXUAL DYSFUNCTION
- OLIGOSPERMIA

INSULIN

DOWN DOSE

KIDNEY DEGRADATION LOST

AVOID

GFR UNDER 50

LACTIC ACIDOSIS

HYPOGLYCEMIA RISK AVOID

MAINSTAY DM + NON-DM CKD

PRURITUS RESISTANT

TREAT HYPERPHOS

MOISTURIZERS

UV LIGHT

KAPPA-OPIOIDS

UROCHROMES = HGB BREAKDOWN PIGMENTS

NFD

NEPHROGENIC FIBROSIS

DEN BARELY DERMOPATHY

NEWER GAD GROUP II OK

IF ESSENTIAL → HD AFTER

AVOID STAGE 4-5

1. Anemia ward — down EPO kidney

WHAT YOU SEE

Sad kidney labeled 'unplugged / down EPO', main cause, starts stage 4.

WHAT IT ENCODES

The peritubular interstitial fibroblasts of the kidney sense hypoxia and make >90% of the body's erythropoietin; as functioning renal mass is lost, EPO production falls and is the MAIN cause of CKD anemia. It is a normocytic, normochromic, hypoproliferative anemia (low reticulocyte count). It typically becomes clinically apparent at eGFR <30 (CKD G3b–G4) and is near-universal by G5/dialysis. Mnemonic: failing kidney = 'unplugged generator' that stops making the EPO signal.

BOARD TRAP

A board stem shows a dialysis patient with Hgb 9, MCV 88, low retic — the wrong answer is to work up B12/folate or call it iron-deficiency microcytosis. CKD anemia is NORMOCYTIC from EPO deficiency. Discriminator: a normal/low MCV with low reticulocytes and otherwise unexplained anemia in a patient with low eGFR points to EPO deficiency, not a nutritional cause — though you still must EXCLUDE iron deficiency before blaming EPO.

2. Iron / functional iron deficiency

WHAT YOU SEE

Iron icons, 'functional iron deficiency', hemolysis, B12/folate.

WHAT IT ENCODES

Two iron problems coexist in CKD: ABSOLUTE iron deficiency (GI blood loss, dialysis circuit losses, poor intake) and FUNCTIONAL iron deficiency (adequate stores but hepcidin-mediated block of iron release from macrophages/enterocytes). Shortened RBC survival (~half-normal, uremic hemolysis) and blood loss compound it. Goals before/with ESA: keep ferritin >100 (non-dialysis) or >200 (dialysis) and transferrin saturation (TSAT) >20–30%. Mnemonic: stores can look 'full' (high ferritin) yet the iron is locked away.

BOARD TRAP

Stem: dialysis patient, ferritin 300, TSAT 15%, not responding to ESA — the wrong move is to withhold iron because 'ferritin is normal.' Functional iron deficiency means iron is trapped despite adequate stores. Discriminator: a LOW TSAT (<20–30%) with a normal-or-high ferritin = functional iron deficiency → give IV iron, do not just escalate ESA. Ferritin is also an acute-phase reactant and falsely elevated by inflammation.

3. Hepcidin / HIF-PHI mechanism

WHAT YOU SEE

Hepcidin and HIF-PHI mechanism icons.

WHAT IT ENCODES

Hepcidin (made by the liver, normally cleared by the kidney) accumulates in CKD and inflammation; it binds ferroportin, internalizing it so iron cannot leave macrophages or be absorbed from the gut — this is the molecular basis of functional iron deficiency. HIF-prolyl-hydroxylase inhibitors (roxadustat, daprodustat, vadadustat) stabilize hypoxia-inducible factor, which lowers hepcidin AND boosts endogenous EPO transcription, raising Hgb orally without injectable ESA. Mnemonic: HIF-PHI 'fakes hypoxia' so the kidney/liver act as if oxygen-starved.

BOARD TRAP

Stem may ask the mechanism of an oral CKD-anemia agent that works without giving exogenous EPO — wrong answer is 'it is a synthetic erythropoietin analog.' HIF-PHIs are NOT ESAs; they stabilize HIF to raise endogenous EPO and suppress hepcidin. Discriminator: an ORAL agent for renal anemia = HIF-PHI; an injectable = epoetin/darbepoetin (ESA). HIF-PHIs carry the same thrombosis caution and should not be used to chase a normal Hgb.

4. Hb target 10–11.5

WHAT YOU SEE

Target/bullseye 'Hb target 10 to 11.5', RIP figure 'don't normalize'.

WHAT IT ENCODES

Treat to a Hgb of roughly 10–11.5 g/dL — enough to relieve symptoms and avoid transfusion, NOT a normal Hgb. KDIGO: start ESA when Hgb <10 and avoid letting it rise above ~11.5. The landmark trials drive this: CHOIR (target 13.5 vs 11.3 → more death/CV events at the higher target), TREAT (darbepoetin to 13 in diabetic CKD → no CV benefit and DOUBLED stroke risk), and Normal Hematocrit Study (hematocrit 42 → more deaths/MI). Mnemonic: aim for the bullseye 'low normal,' not the red zone.

BOARD TRAP

Stem: a provider wants to push ESA to normalize Hgb to 13 to make the patient 'feel best' — wrong answer is to agree. Targeting normal/high Hgb with ESAs increases stroke, thrombosis, vascular access loss, and mortality (CHOIR, TREAT, Normal Hct). Discriminator: the correct ceiling is ~11.5 g/dL; the moment a stem mentions targeting 12–13+ with an ESA, that is the trap.

5. Anemia Rx — ESA / iron / HIF-PHI

WHAT YOU SEE

Shelf of bottles: ESA, iron, B12/folate, HIF-PHI.

WHAT IT ENCODES

Sequence: (1) exclude/treat OTHER causes (iron, B12, folate, blood loss); (2) replete IRON FIRST to ferritin/TSAT goals (oral often fails — IV iron, e.g., iron sucrose or ferric carboxymaltose, preferred in dialysis); (3) then start an ESA — epoetin alfa ~50–100 units/kg SC/IV up to 3x/week, or darbepoetin alfa (longer half-life, dosed weekly–monthly), OR an oral HIF-PHI. Always confirm adequate iron BEFORE and DURING ESA therapy. Mnemonic: 'fuel before the engine' — iron before EPO.

BOARD TRAP

Stem: anemic dialysis patient with TSAT 14%/ferritin 90 is started straight on epoetin and fails to respond — wrong answer is to keep escalating the ESA dose. ESA without adequate iron fails (functional/absolute iron deficiency = ESA hyporesponsiveness). Discriminator: the next best step is IV iron, not more ESA. Other causes of ESA resistance to remember: ongoing inflammation/infection, hyperparathyroidism, aluminum, and underdialysis.

6. Uremic bleeding patient

WHAT YOU SEE

Bruised patient in bed 'uremic bleeding', down platelet aggregation, long bleeding time.

WHAT IT ENCODES

Uremic toxins impair platelet FUNCTION: defective platelet–vessel-wall adhesion and aggregation, partly via abnormal vWF/GPIIb interaction, increased nitric oxide/prostacyclin, and anemia (RBCs normally push platelets toward the vessel wall). Result: mucocutaneous bleeding (GI, ecchymoses, oozing) with a PROLONGED bleeding time but NORMAL platelet count, PT, and PTT. Mnemonic: the platelets are present but 'drunk on urea.'

BOARD TRAP

Stem: uremic patient with GI bleed, platelets 220k, normal PT/PTT — wrong answer is to transfuse platelets or call it thrombocytopenia. Uremic bleeding is QUALITATIVE platelet dysfunction; the count is normal and platelet transfusions don't fix it (transfused platelets become dysfunctional too). Discriminator: prolonged bleeding time with normal count/PT/PTT = uremic platelet dysfunction → treat with dialysis/DDAVP, not platelet units.

7. DDAVP / cryo / estrogens / TXA

WHAT YOU SEE

Bottles 'first-line acute = DDAVP', cryoprecipitate (vWF+factor VIII), estrogens, TXA.

WHAT IT ENCODES

Acute uremic bleeding ladder: DDAVP (desmopressin) 0.3 mcg/kg IV/SC is first-line — releases vWF/factor VIII from endothelium, works in ~1 hour but tachyphylaxis after 2–3 doses; cryoprecipitate gives vWF + factor VIII for a longer effect; conjugated estrogens (~0.6 mg/kg IV daily x5) give a slow but durable effect (days–weeks). Correcting anemia (raise Hct toward ~30%) and DIALYSIS are also therapeutic; TXA is an antifibrinolytic adjunct. Mnemonic: 'Dialysis Definitively Cures, DDAVP Does it Fast.'

BOARD TRAP

Stem: dialysis patient with refractory uremic mucosal bleeding asks for the DEFINITIVE fix — wrong answer is DDAVP alone. DDAVP is rapid but short-lived (tachyphylaxis); the definitive correction of uremic platelet dysfunction is intensified DIALYSIS (clearing the uremic toxins). Discriminator: 'fastest' = DDAVP; 'most definitive/long-term' = dialysis; 'durable bridge' = cryoprecipitate or conjugated estrogens.

8. Hypercoagulable / nephrotic

WHAT YOU SEE

Spider/clot monster 'hypercoagulable', nephrotic syndrome, lost antithrombotic proteins.

WHAT IT ENCODES

Nephrotic-range proteinuria (>3.5 g/day) creates a HYPERCOAGULABLE state: urinary loss of antithrombin III (plus protein S/plasminogen) and compensatory hepatic synthesis of procoagulants (fibrinogen, factors V/VIII). Highest risk with membranous nephropathy and albumin <2.5–2.8 g/dL. Classic complication: renal vein thrombosis (flank pain, hematuria, sudden creatinine rise/LDH bump) and other VTE/PE. Mnemonic: you 'pee out the brakes' (antithrombin III) so clotting runs away.

BOARD TRAP

Stem: CKD = bleeding risk, so the wrong instinct is to assume all renal patients only bleed. CKD is PARADOXICAL — uremia bleeds, but nephrotic syndrome CLOTS. Discriminator: heavy proteinuria + low albumin + acute flank pain/hematuria/AKI = renal vein thrombosis from antithrombin-III loss; the answer is anticoagulation, not a bleeding workup. Membranous nephropathy is the classic nephrotic cause linked to RVT.

9. Anticoagulation shelf

WHAT YOU SEE

Shelf 'anticoagulation': avoid warfarin (calciphylaxis), DOAC clearance caution, UFH/LMWH notes.

WHAT IT ENCODES

Anticoagulation in CKD is a tightrope (coexisting bleeding AND clotting risk). Warfarin promotes vascular/calciphylaxis calcification (inhibits matrix Gla protein) and is hard to dose; DOACs are renally cleared (dabigatran ~80% renal — avoid in advanced CKD; apixaban is the least renally dependent and generally preferred). LMWH accumulates with low eGFR (anti-Xa monitoring or dose reduction; avoid if CrCl <30); unfractionated heparin (hepatic/reticuloendothelial clearance) is the safest IV choice in severe CKD/ESRD. Mnemonic: 'when kidneys fail, fall back to heparin.'

BOARD TRAP

Stem: ESRD patient on hemodialysis needs full therapeutic anticoagulation for a DVT — wrong answer is therapeutic LMWH (enoxaparin) at standard dose. LMWH is renally cleared and accumulates → bleeding when CrCl <30. Discriminator: in advanced CKD/ESRD use UNFRACTIONATED heparin (non-renal clearance, easily reversed) or carefully dosed apixaban; avoid warfarin where calciphylaxis is a concern and avoid dabigatran. A modest creatinine context still demands eGFR-based DOAC dosing.

10. Neuromuscular ward / peripheral neuropathy

WHAT YOU SEE

Brain + legs 'peripheral neuropathy', restless legs, asterixis, seizures, 'CNS stage 3 onwards'.

WHAT IT ENCODES

Uremic neurotoxicity spans peripheral and central nervous systems: a distal, symmetric, sensorimotor polyneuropathy (length-dependent, legs first); autonomic dysfunction; and CENTRAL uremic encephalopathy with cognitive slowing, asterixis, myoclonus, and seizures. CNS findings generally emerge as eGFR falls (stage 3 onward, worst in G5). Mnemonic: 'flapping hands and twitchy legs' = the brain is marinating in urea.

BOARD TRAP

Stem: advanced-CKD patient develops confusion, asterixis, and a seizure — wrong answer is to load antiepileptics and image the brain as the primary fix. Uremic encephalopathy/seizures is a recognized INDICATION FOR DIALYSIS (the AEIOU mnemonic: Acidosis, Electrolytes/hyperK, Intoxication, Overload, Uremic symptoms — pericarditis, encephalopathy, bleeding). Discriminator: encephalopathy/asterixis/seizure in uremia → start dialysis; symptoms reverse with clearance.

11. Restless legs / dialysis indication

WHAT YOU SEE

Figure 'restless legs relieved by movement', 'dialysis indication'.

WHAT IT ENCODES

Restless legs syndrome (uncomfortable urge to move legs, worse at rest/evening, RELIEVED by movement) is common in CKD and worsened by iron deficiency. Manage by correcting iron (ferritin goal often >100), optimizing dialysis, and dopamine agonists (e.g., pramipexole) or gabapentinoids; severe refractory uremic neurologic disease supports dialysis initiation. Mnemonic: legs that 'must keep walking' at night.

BOARD TRAP

Stem: dialysis patient with classic RLS and ferritin 40 — wrong answer is to jump straight to a dopamine agonist. Discriminator: in CKD/RLS, CHECK AND REPLETE IRON FIRST (low ferritin is a reversible driver of RLS) before pharmacologic dopaminergic therapy; also review for offending drugs (antihistamines, antidopaminergic antiemetics).

12. Endocrine apothecary — drug dosing

WHAT YOU SEE

Shelf with insulin, 'down dose / kidney degradation lost', 'avoid hypoglycemia'.

WHAT IT ENCODES

The kidney both clears and degrades insulin; as eGFR falls, insulin (and many sulfonylureas/their metabolites) accumulate, so insulin requirements DECREASE and hypoglycemia risk rises. Practical board points: reduce insulin doses as CKD advances; avoid glyburide (renally cleared active metabolite → hypoglycemia) and prefer glipizide or shorter agents; metformin contraindicated when eGFR <30 (lactic acidosis risk). Mnemonic: failing kidneys 'hoard insulin' → less is needed.

BOARD TRAP

Stem: long-standing diabetic with worsening CKD now has recurrent hypoglycemia on a stable insulin/glyburide regimen — wrong answer is to add carbohydrate or call it 'good control.' Discriminator: reduced renal insulin/sulfonylurea clearance is the cause → DECREASE insulin and STOP glyburide (switch to glipizide). Failing to down-titrate as eGFR drops causes the hypoglycemia.

13. SGLT2 / GLP-1 dialysis notes

WHAT YOU SEE

Bottles 'SGLT2' (avoid at low GFR), GLP-1, women menstrual/fertility note.

WHAT IT ENCODES

SGLT2 inhibitors LOSE glucose-lowering efficacy at low eGFR (they need filtered glucose to block), but their RENAL and CV protection persists down to low thresholds (e.g., dapagliflozin/empagliflozin continued to ~eGFR 20–25, even in non-diabetic proteinuric CKD per DAPA-CKD/EMPA-KIDNEY) — they are continued for organ protection, not sugar. GLP-1 agonists aid weight/glycemia and have CV/renal benefit. CKD also causes hypothalamic-pituitary-gonadal dysfunction: women get menstrual irregularity/amenorrhea and reduced fertility; men get hypogonadism/ED. Mnemonic: SGLT2i = 'protect the organ even when it stops lowering sugar.'

BOARD TRAP

Stem: proteinuric CKD patient at eGFR 28 — wrong answer is to STOP the SGLT2 inhibitor because 'it no longer lowers glucose at this GFR.' Discriminator: SGLT2i are continued for renal/CV protection well into advanced CKD (and benefit non-diabetics); their flagging glycemic effect at low eGFR is NOT a reason to discontinue. Also do not assume infertility in CKD women is permanent — gonadal dysfunction can improve after transplant.

14. Pruritus / NFD dermatology corner

WHAT YOU SEE

Itchy figure 'pruritus resistant', urochromes, 'NFD = nephrogenic fibrosing dermopathy / avoid gadolinium'.

WHAT IT ENCODES

Uremic (CKD-associated) pruritus is often intense and treatment-resistant; linked to uremic toxins, hyperphosphatemia, secondary hyperparathyroidism, and dysregulated opioid receptors — manage by optimizing dialysis/phosphate, emollients, gabapentinoids, UVB, and difelikefalin (kappa-opioid agonist). Skin also shows sallow yellow-brown urochrome pigmentation. Gadolinium-based MRI contrast in advanced CKD can trigger nephrogenic systemic fibrosis (NSF, formerly nephrogenic fibrosing dermopathy) — fibrosis of skin and organs. Mnemonic: 'gadolinium + dead kidneys = stiff fibrotic skin.'

BOARD TRAP

Stem: patient with eGFR 22 needs an MRI and the team plans gadolinium contrast — wrong answer is to proceed routinely. Discriminator: AVOID gadolinium (or use newer macrocyclic agents only with caution/lowest dose) when eGFR <30 or on dialysis due to NSF risk; this parallels avoiding IODINATED contrast for contrast-associated AKI, but the GADOLINIUM danger is FIBROSIS (NSF), not AKI — don't confuse the two contrast hazards.